Linking

Learning more about an individual or a group by associating data items or user actions. Linking may lead to unwanted privacy implications, even if it does not reveal one's identity.

L.1 Linked data

Linking through identifiers

L.1.1 Unique identifier

Linking based on an identifier that is unique (globally or locally, within the system or across the context boundary)

Examples

Email address as ID: Using an email address as an identifier enables the linking of all activities to the same individual, even across multiple services.

IP addresses: An IP address can be used to link multiple visits to the same individual.

Logged in web shop: All product views in a web shop are linked to the same user because they are logged in.

User IDs: User IDs can be used to link different requests to the same user.

Criteria

Is there an identifier (unique in system/session) or dataset?

Is there other data associated with that identifier?

Is there previous data with the same identifier to which new data can be linked?

Impact

Construct profile: Unique identifiers facilitate the linking of new data items to a user profile, accumulating growing amounts of personal data associated with this profile. This can later lead to 'Identifying' threats.

Additional Info

Authenticated users: Linking is especially easy for authenticated users, as all requests in the same session are linked.

L.2 Linkable data

Linking through the combination or analysis of data

L.2.1 Through Combination

Linking through combination

L.2.1.1 Quasi-identifier combining data of a single individual

Linking through a set of attributes that can serve as a quasi-identifier

Examples

Location trace: A small set of locations can be used to uniquely link activity to a single user.

Age-gender-postalcode: A subset of attributes may be sufficient to uniquely link data to a particular individual.

Browser fingerprint: A browser fingerprint combines properties (OS, browser, display size, ...) that together are unique to a website visitor.

Fine-grained raw data: Fine-grained raw data is stored that enables linking specific entries to individuals.

Criteria

Is there a set of attributes that can serve as an identifier?

Is there other data sent together with that quasi-identifier?

Is there existing data to link it to?

Impact

Construct profile: The use of quasi-identifiers enables the linking of new data items to a user profile to gather increasing amounts of personal data, even without unique identifiers.

Additional Info

Combined properties: Many requests contain a lot of different properties that, when combined, are unique to an individual.

L.2.1.2 Combining data of different individuals

Linking through combining data

Examples

Social network indirect relations: Relations in a social network indirectly contain information about individuals not on that social network.

Contact lists: Contact lists that are uploaded to services allow service providers to indirectly reconstruct the social graph of the involved individuals.

Criteria

Is data from multiple users combined?

Are datasets or records combined?

Impact

Construct profile: The combination of records from multiple different sources enables building a more comprehensive profile on an individual.

Additional Info

Data subject different from sender: The data subject is not necessarily the sender of the data.

Quasi-identifiers: Quasi-identifiers can be sufficient to link and combine data from multiple users.

L.2.2 Through Profiling, derivation, or inference

Linking through derivation or inference

L.2.2.1 Profiling an individual

Personal data of an individual can be analyzed for patterns to link data to that individual. Association is performed at the basis of the data or user actions.

Examples

Facial expression analysis: By applying sentiment analysis to faces in pictures, the emotional state of an individual can be derived.

Telemedicine e-health system: The frequency of data exchanges from a health monitoring device allows an adversary to infer a patient's medical condition.

Requests to external services: External service requests may be linked to the same user by inferring they are sent in the context of an ongoing session.

Personalized shopping recommendations: Shopping recommendations sent to consumers are based on the degree of similarity to other consumers.

Criteria

Are there patterns derivable from the data?

Can (new) personal data be inferred from the linked data points?

Impact

Link additional data: Deriving patterns from the data can facilitate the linking of data that was not intended to be linked.

Link requests: Timing patterns of messages can be used to link requests to construct profiles.

Indirectly reveal sensitive data: Indirect links can reveal sensitive data about individuals that are not involved in the original dataset.

Additional Info

Service usage: The more data is collected and the more detailed it is, the easier it can become to discern patterns for linking.

L.2.2.2 Profiling a group of individuals

Data of a single individual leads to insights about a larger group of individuals. Knowledge transfers to other individuals through known relations of group membership or similarity.

Examples

DNA testing service: Findings obtained through DNA analysis (e.g., genetic predisposition for certain illnesses) by an individual may also apply to their family members.

Energy consumption meter: Readings obtained from an energy consumption meter do not apply to an individual but to households. Consumption patterns obtained for an individual (e.g. heating cost) may be generalized.

Criteria

Is there data being analyzed for patterns to generalize?

L.2.2.3 Profiling an individual through (dis)similarity

Applying methods to assess uniqueness or similarity to predict relevant properties.

Examples

Salary statistics queries: Querying the average salary with a strict set of criteria can reveal the salary of an individual employee.

Through writing style: Patterns in a person's writing style can be used to link multiple texts written by the same person.

Analyzing timing patterns: Unique timing patterns could be recognized to link the use of a service to the same user.

Analyzing messages: Messages can be analyzed for patterns (e.g., writing style, timing) to link them to the same user.

Criteria

Is the data from different individuals distinguishable?

Is it possible to query data using distinguishable attributes?

Impact

Construct profile: When users are distinguishable from one another, user profiles can be constructed to collect growing amounts of personal data associated with each profile. This can later lead to 'Identifying' threats.

Additional Info

Data without attributes: Even data without clear attributes could contain patterns to allow distinguishing between different individuals.